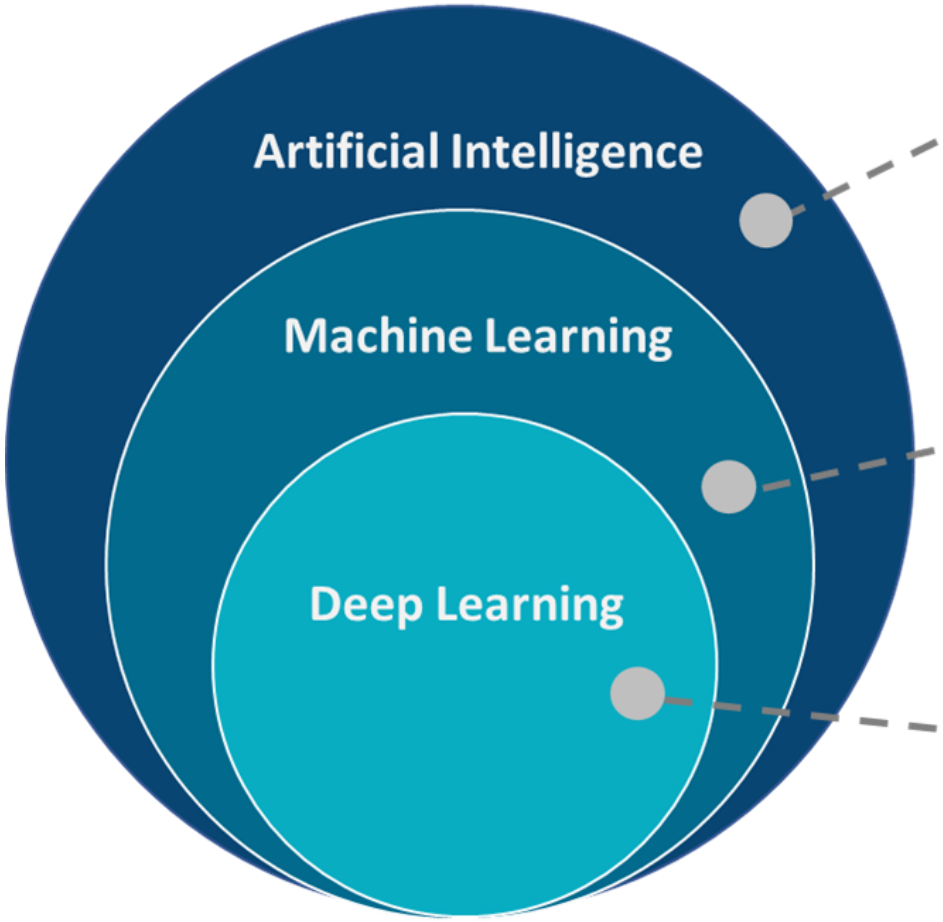
Deep Learning Architectures

https://developer.ibm.com/articles/cc-machine-learning-deep-learning-architectures/

EE/CS 228 – Yinglun Zhu

What is AI? ML? DL?

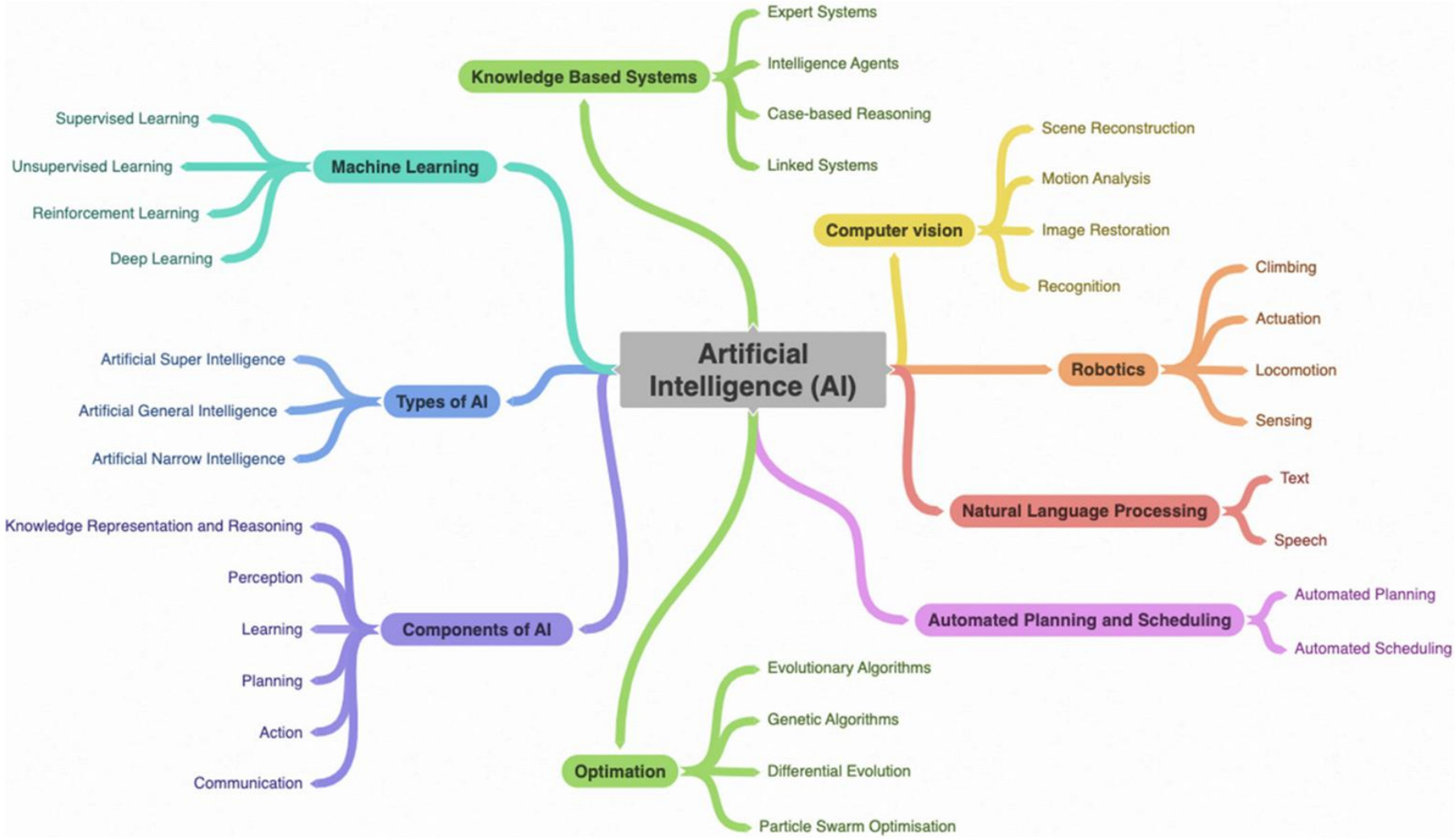


Artificial Intelligence
A technique which enables machines to mimic human behaviour

Machine Learning
Subset of AI technique which uses statistical methods to enable machines to improve with experience.

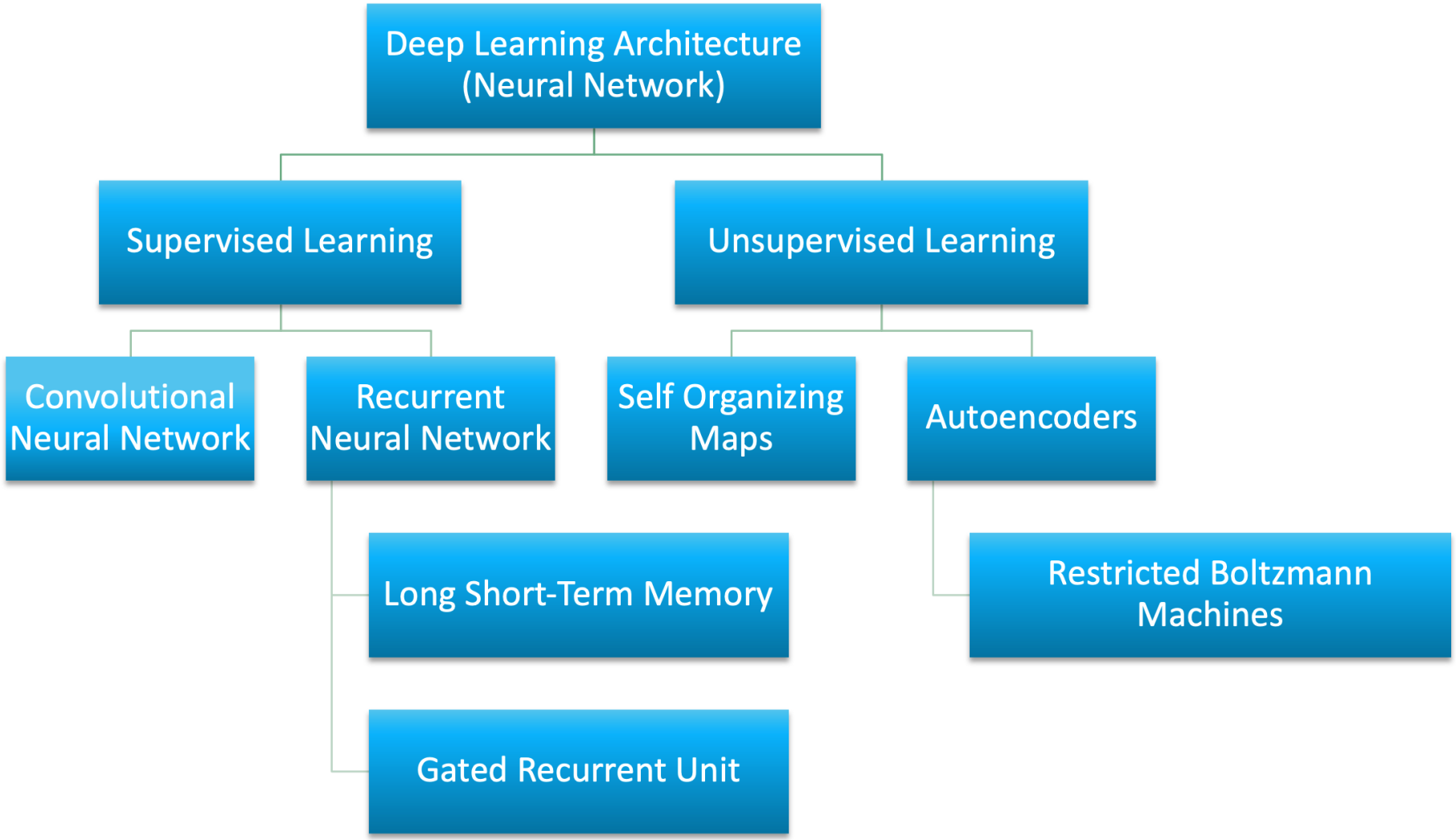
Deep Learning
Subset of Machine Learning which make the computation of multi-layer neural network feasible

Overview of AI

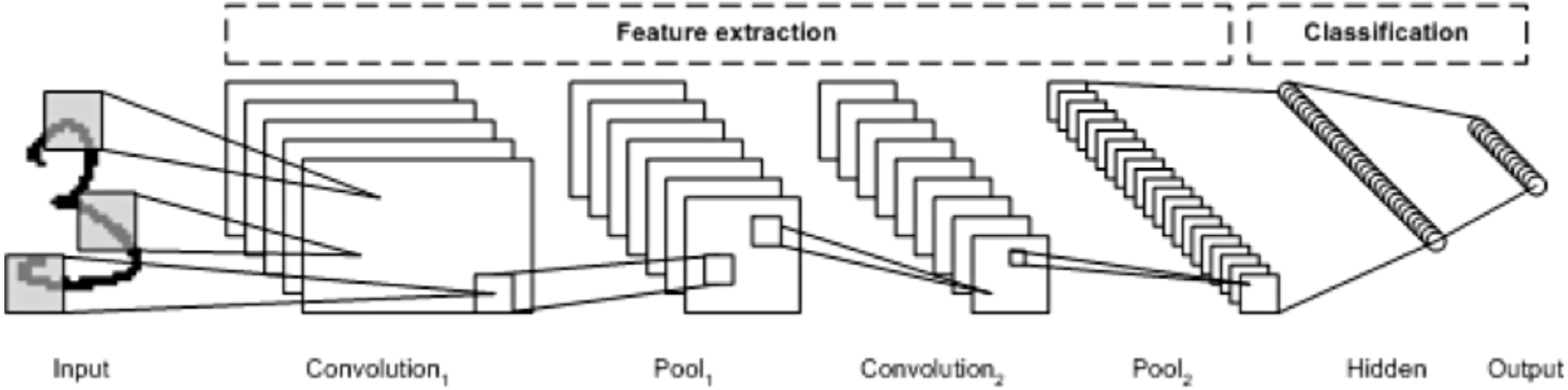


https://www.semanticscholar.org/paper/Opportunities-and-Adoption-Challenges-of-AI-in-the-Regona-Yigitcanlar/fe7c42d5e0e73769e35903f542d0d98020e7511c

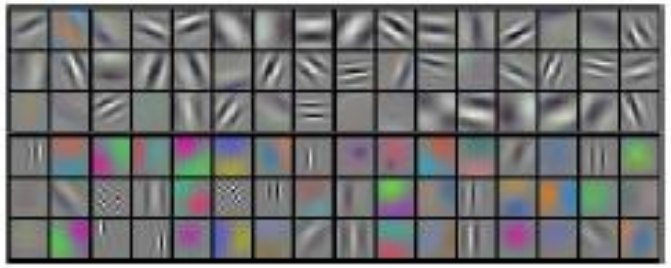
Hierarchy of Learning Approaches



Convolutional Neural Networks

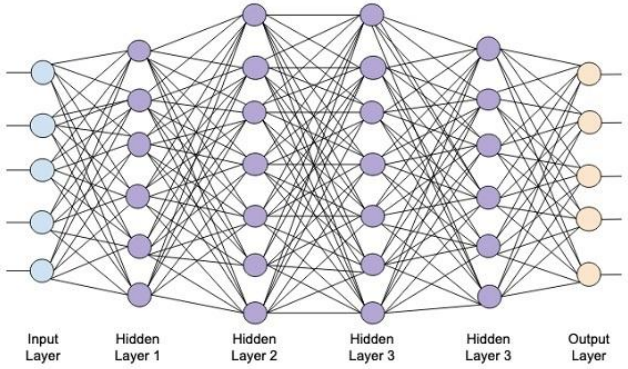


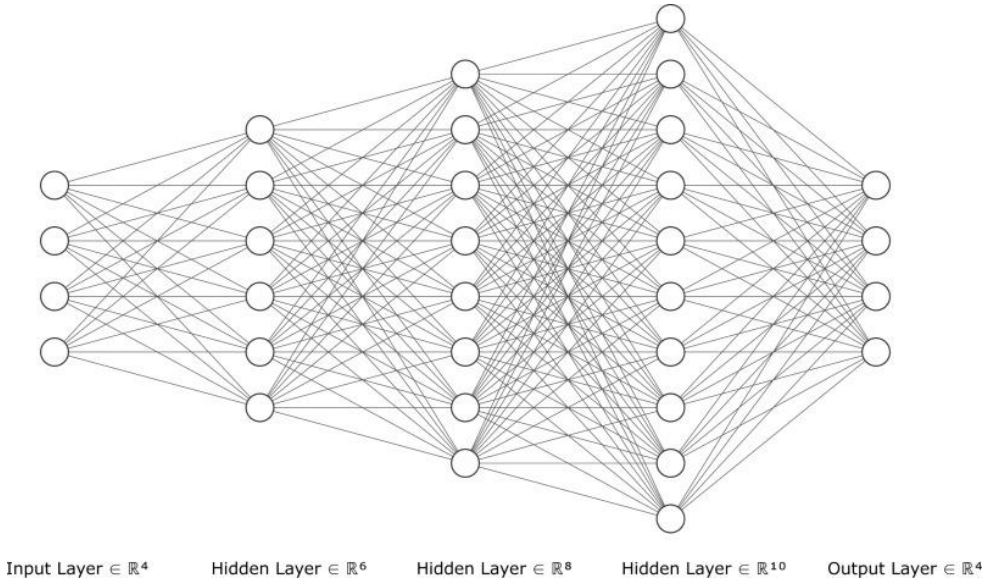
Convolution, Pooling, Dropout



96 convolutional kernels of size 11113 learned by the first convolutional layer on the 2242243 input images. – AlexNet, NeurIPS 2012

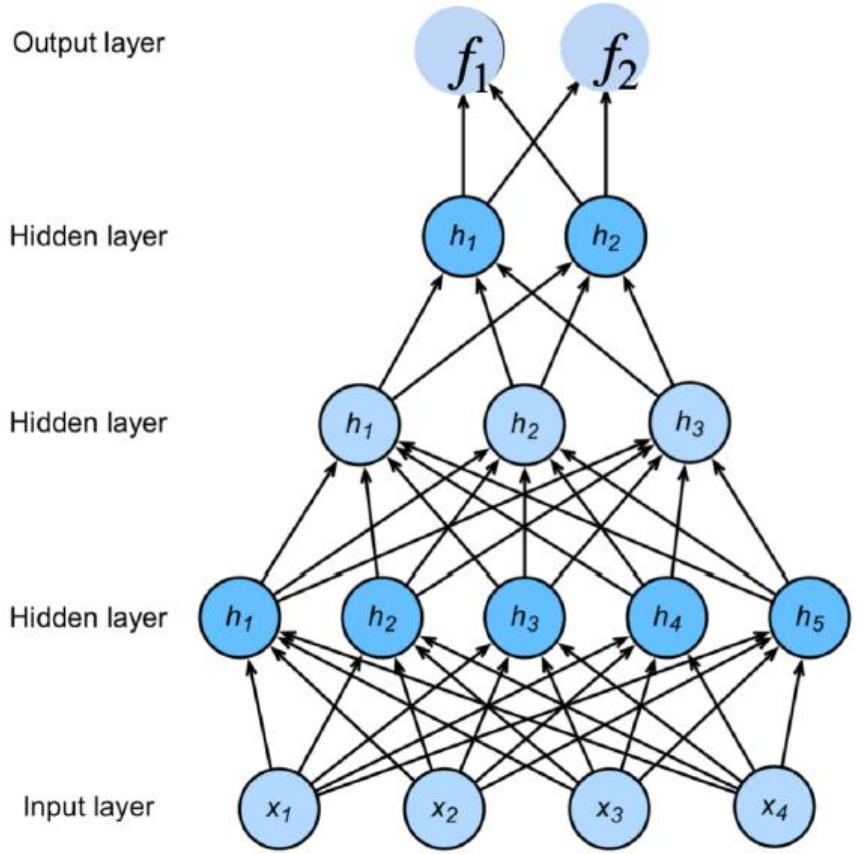
Larger Networks?



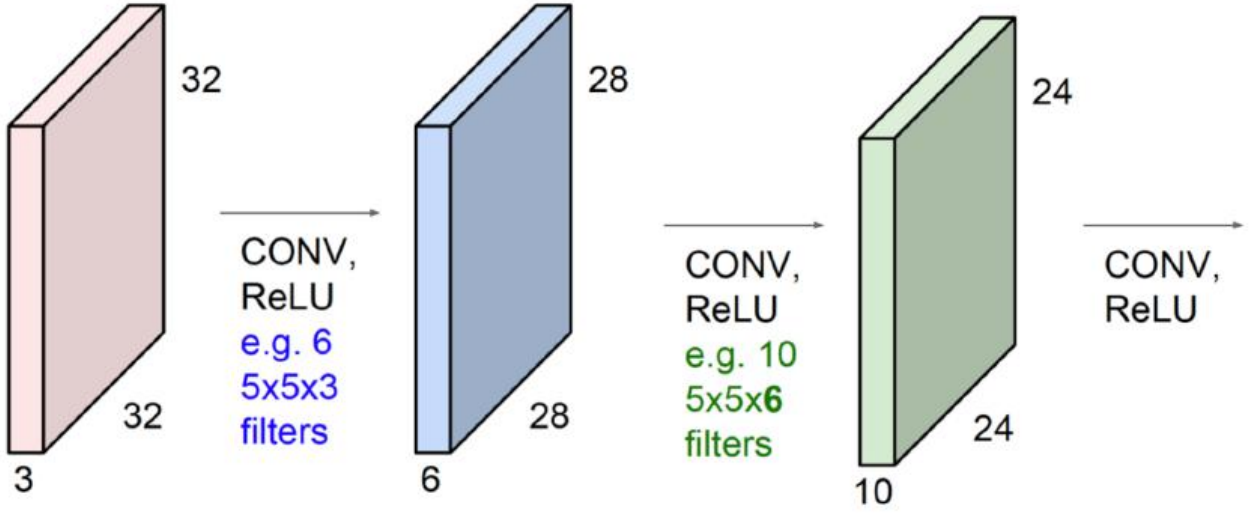


- The larger the better???
- No overfitting (layman term – memorize training data only, poor performance when testing data is used)
- What the reason of overfitting??? One of the reason is we have too many parameters

Feed-forward Networks

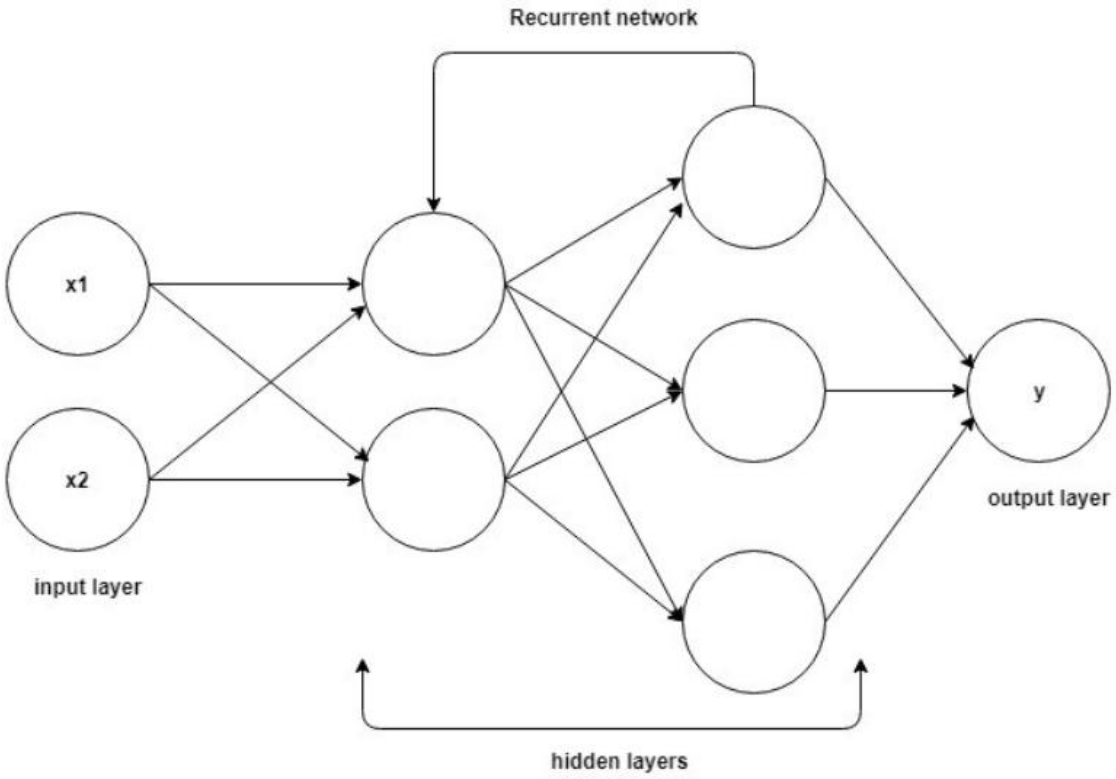


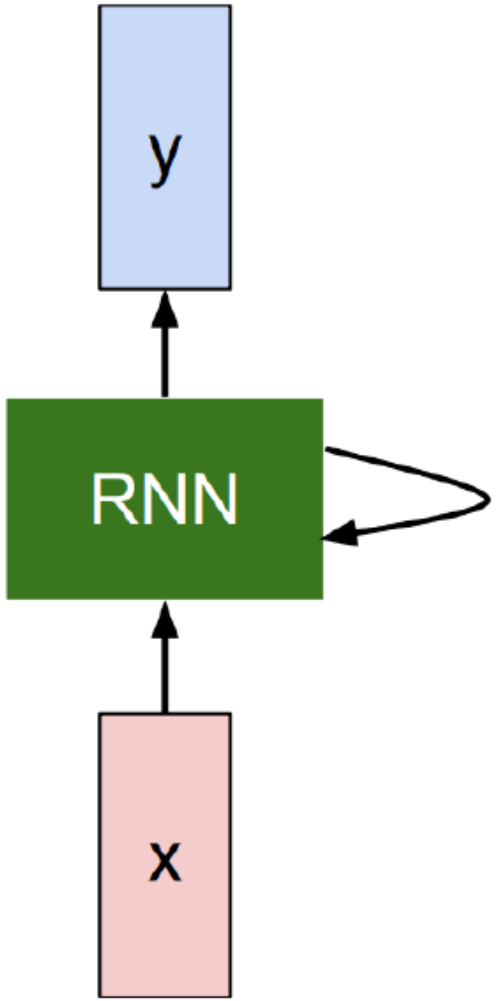
MLP



CNN

Recurrent Neural Networks





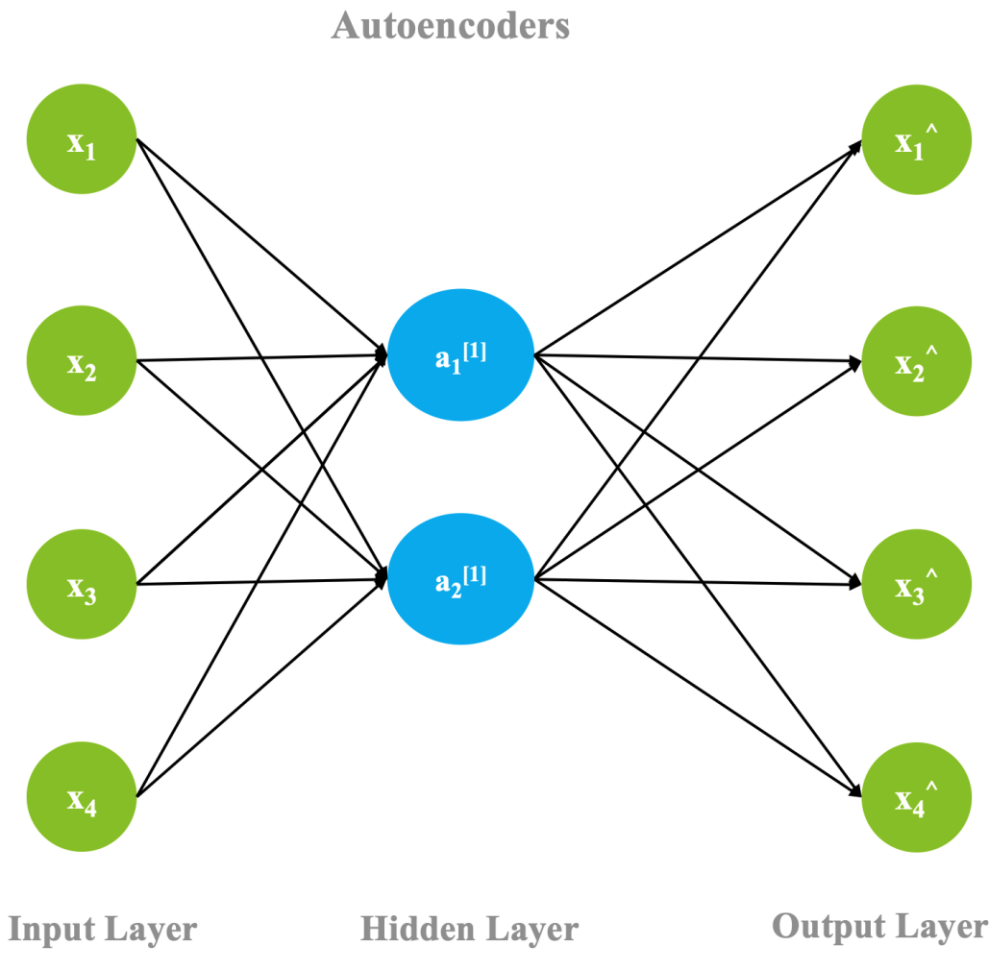
$$\boxed{h_t} = \boxed{f_W}(\boxed{h_{t-1}}, \boxed{x_t})$$

new state

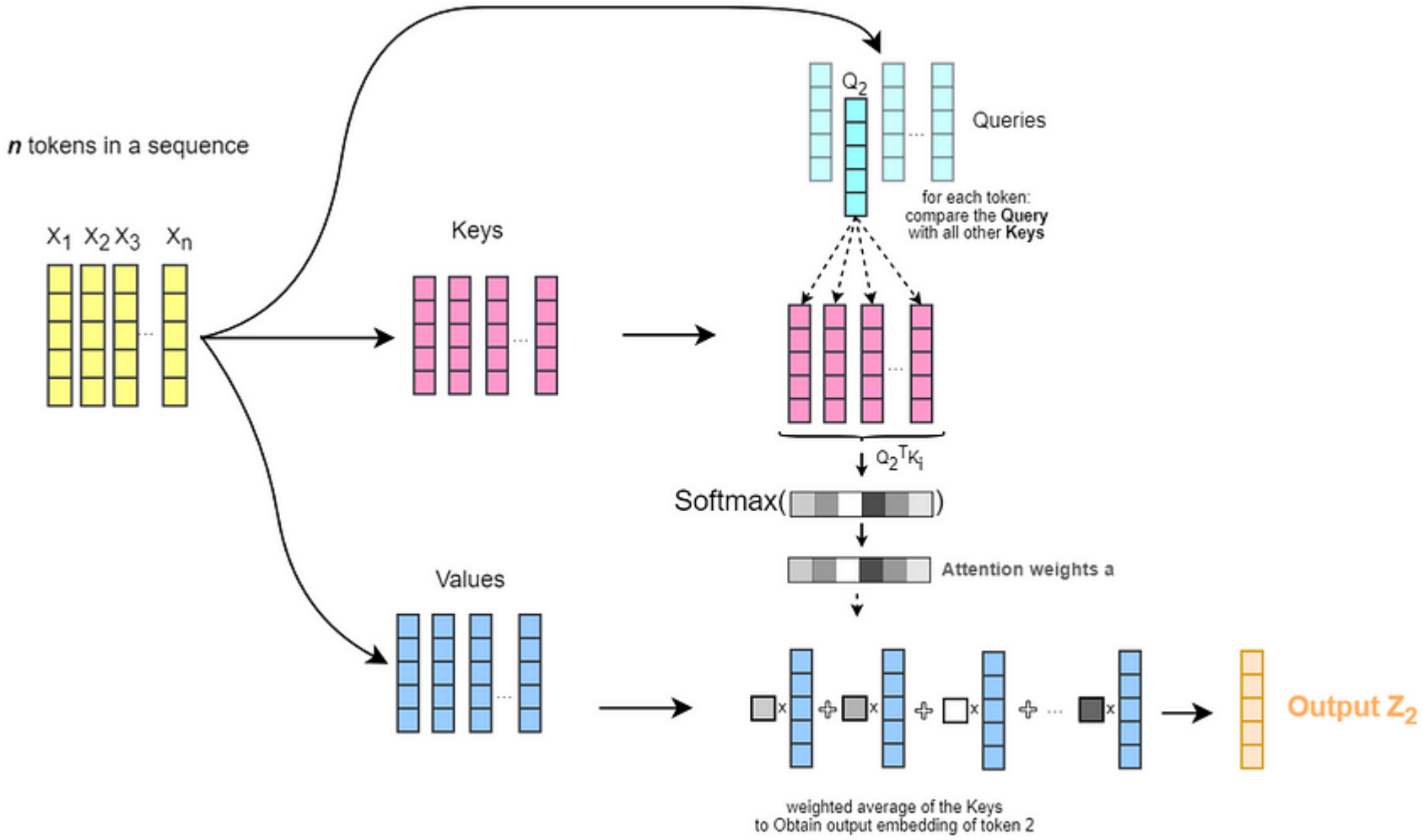
old state

input at some time step

Autoencoders

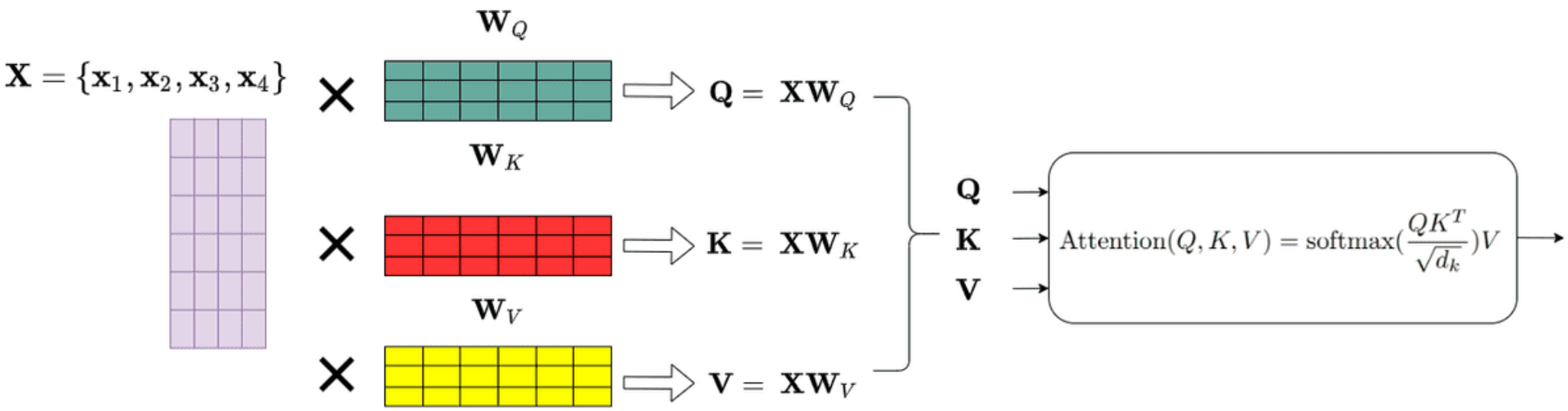


Transformers – Query, Key, Value

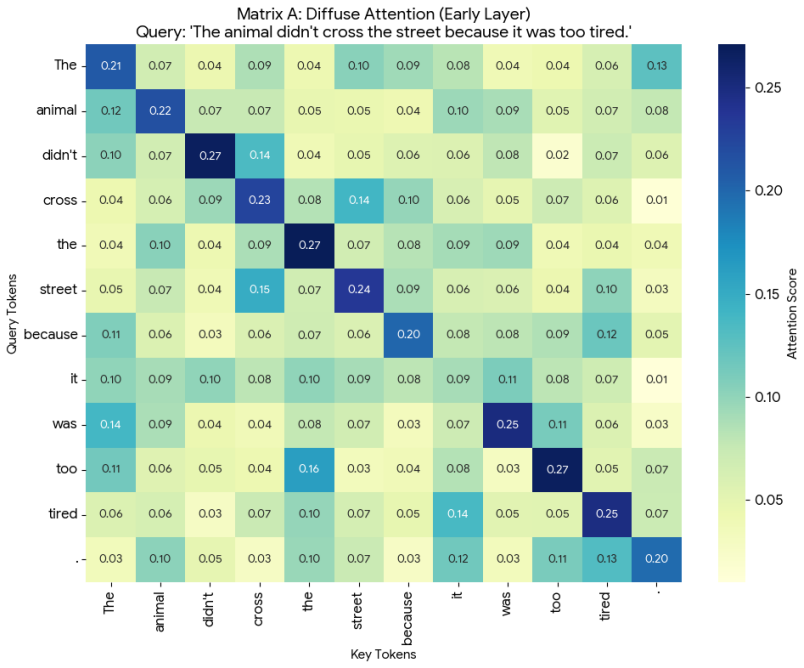


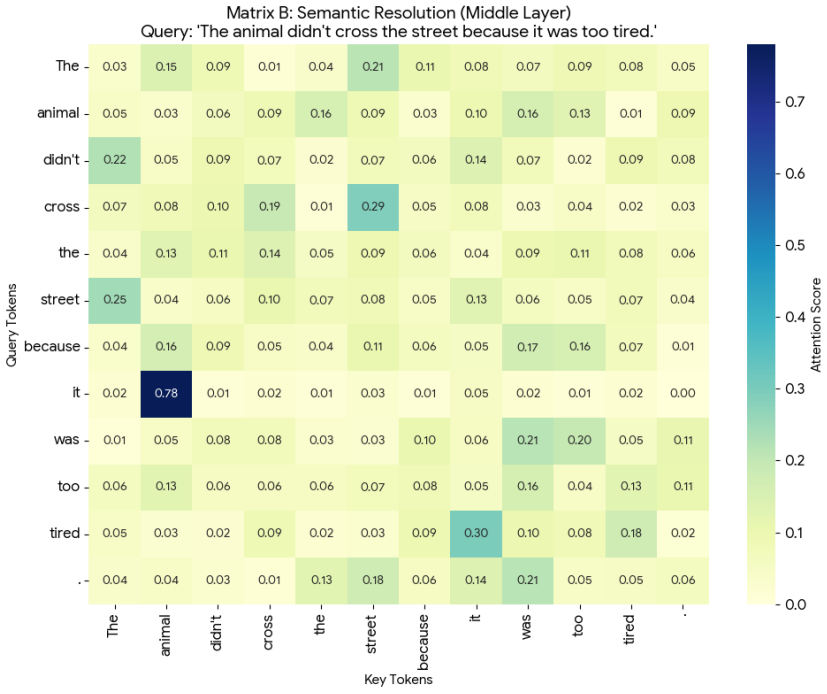
https://towardsdatascience.com/what-are-query-key-and-value-in-the-transformer-architecture-and-why-are-they-used-acbe73f731f2

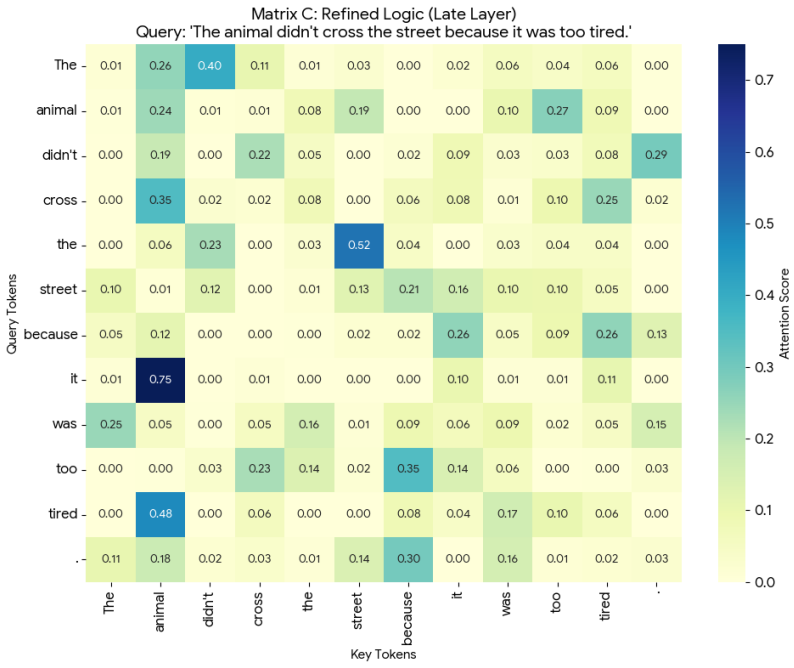
Self-Attention



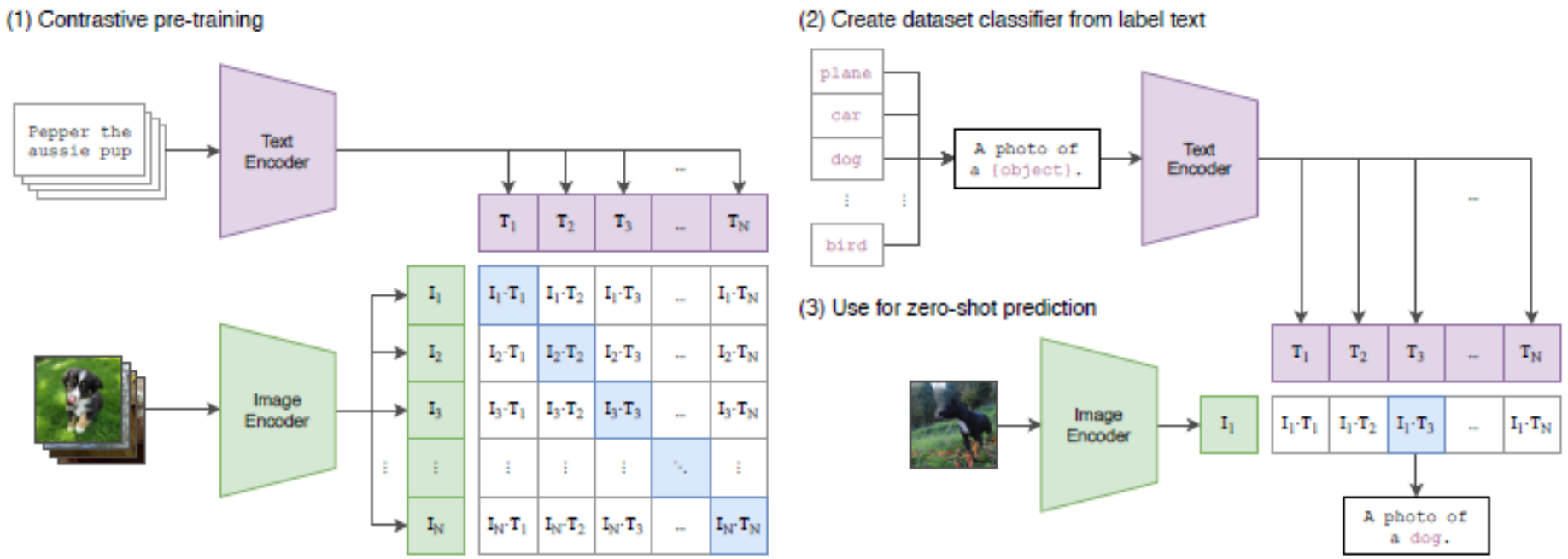
Attention Matrices







Vision-Language Representation



Advantages and Issues with LLMs/VLMs

- Zero shot recognition
- Catastrophic forgetting
- Hallucinations
- Security